

3.5 Experiments, Results and Analysis

In this section, we present several experiments focusing on HRTF personalization efforts and are presenting the outcomes herein. Our methodology aligns with and extends the work detailed in "Head Related Impulse Response Interpolation and Extrapolation Using Deep Belief Networks," [65] authored by G. Kestler, Shahrokh Yadegari, and D. Nahamoo, published in 2019. This study, discussed previously (see Section 3.4.2), introduced a method employing a stack of fully connected neural networks for the interpolation and extrapolation of HRTF databases using existing HRTF data alongside anthropometric measurements of CIPIC subjects. Our investigation explored the potential of advancing this method to generate HRTFs for any spatial position around the head, solely based on an individual's anthropometric data.

Throughout the research and adaptation of the method proposed in the past study to personalisation task, we encountered an overfitting problem due to small size and insufficiency of the CIPIC data. We varied training parameters such as the learning rate and its scheduling and we experimented with different amounts of dropouts. We also tested the effects of masking and weighting a loss function to guide the model's focus towards certain spatial areas. We explored and tested the efficiency of the original paper's method, which involves intermediate steps and stacking trained networks. We also tried to clean and correct the CIPIC dataset, which lacks certain anthropometric data, and to mitigate error artifacts we think are result of recordings process inefficiencies. We explored the possibilities of improving model metrics through this approach by training a model on the adjusted data.

The subsequent sections detail our experiments, providing quantitative results and visual representations, especially focusing on LSD numbers and graphs, together with synthesized HRTFs for some meaningful data points across azimuth and elevation for both training and testing subjects. Our report will highlight findings that either showed some meaningful results or offered insightful revelations regarding HRTF generation challenge.

3.5.1 Base Model

The first experiment we conducted involved training existing models from [65] on full CIPIC dataset without excluding any positions, allowing the network to learn the full space of the data for each subject. We utilized all 35 subjects with available anthropometric data from CIPIC and divided the data into training, validation, and test sets with proportions of 70%, 20%, and 10% respectively (resulting in 4 test subjects). We

trained all networks for 20 epochs and 20 it-

erations, mixing validation data and removing new subjects as validation data as described in the original paper, except for the "magl" and "magr" networks, which we trained for 25 epochs and 25 iterations. We then evaluated whether the network could successfully predict HRTF curves for test subjects it had never seen in the time of training, given anthropometric data and a set of positions. The learning rate for the experiment was set to 0.0005.

The Table 3.5.1 shows LSD numbers for the experiment, for both left and right ears across two different frequency ranges: the full spectrum and below 11kHz.

The Figure 3.5.1 shows LSD graphs for training subject 003 and one of the test subject 009. It is evident that the results for the

test subject are not favorable, while the training subject shows a overly good approximation. This is a clear indication of overfitting, which is expected for such a small dataset as CIPIC.

To provide further insight into the model's performance, Figure 3.5.2 showcases recon-

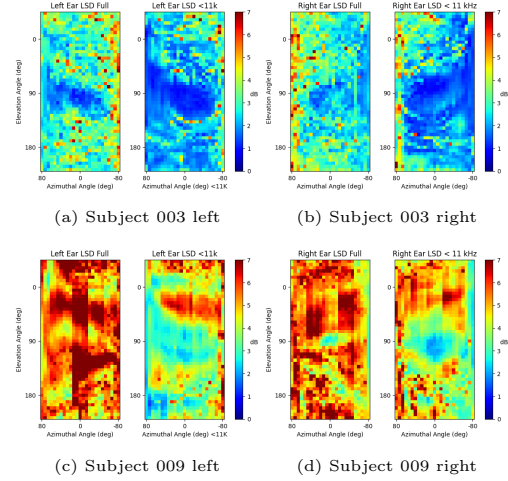


Figure 3.5.1: HRTF LSD images for all positions for left and right ear subject 003 from training set and 009 from test set.

Subject	Ear	Full Spectrum	Below 11kHz
Test	Left	5.844	3.983
	Right	5.489	4.402
Training	Left	3.138	2.275
	Right	3.034	2.353
Test zero elevation	Left	5.238	4.979
	Right	5.332	3.956

Table 3.5.1: LSD Results for base model

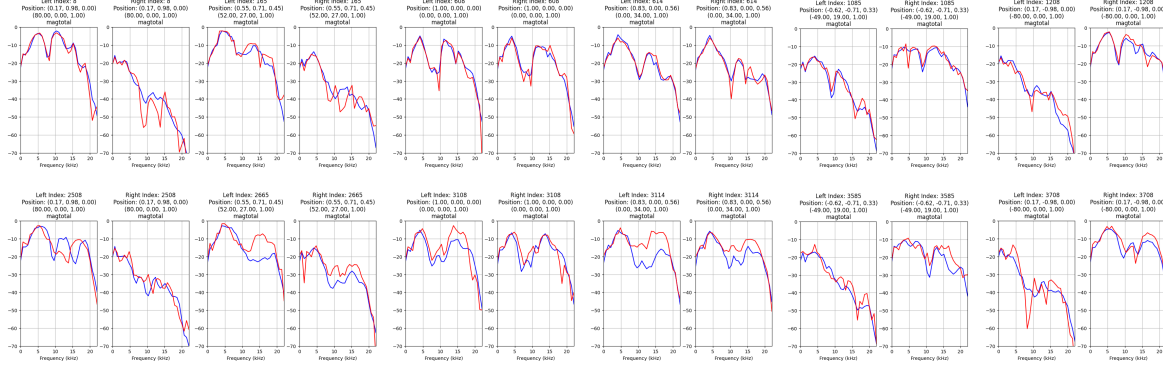


Figure 3.5.2: HRTF LSD images for some important positions for the left and right ears of subjects 003 from the training set (upper row) and 009 from the test set (lower row).

structions for several crucial data points. From left to right, the depicted points include the leftmost position at azimuth zero, the left halfway point with elevation, the center with no elevation, and the center with elevation. Similarly, reconstructions for the corresponding points on the right side are also displayed. These points are presented for training subjects 003 on the upper row and for the test subject 009 on the lower row. While reconstructions for both subjects show relatively meaningful matches to real HRTF curves, those for the test subject often exhibit mismatches and training shows too precise approximation, again indicating a potential overfitting problem. This discrepancy highlights the importance of robust model evaluation and the need for techniques to mitigate overfitting in our models.

3.5.2 Dropouts

Continuing our research, we tried dropout regularization as a remedy to the overfitting problem. By introducing dropout regularization, we aim to improve the generalization capabilities of our model and reduce overfitting tendencies. Accordingly, we attempted to address this issue by implementing various dropout levels into our training process. We kept all the parameters of the model, data, and training procedure constant, same as in the baseline. First, we experimented with dropout rate of 10%. The results of the experiments are summarized in the table table 3.5.3 and corresponding LSD figures are shown in the figure 3.5.4.

As illustrated by the data presented in the table and figure, the model exhibits a marginal improvement in the test LSD and slightly worse results for the training subjects, which might be interpreted as a reduction in overfitting. However, upon closer examination, most generated HRTFs appear similar across data points, and the reconstruction reveals that the model is clearly underfitting the data. In Figure 3.5.5, the predictions for both the test and training subjects, as well as between different points, predominantly reflect the mere averages of the curves, offering only general trends. Such outcomes cannot be deemed satisfactory for our research objectives. This issue mirrors the problems reported in the work of Miccini et al. [106], highlighting a recurring challenge in accurately modeling HRTFs.

We also experimented with various dropout rates, encountering issues similar to those described above. Although these adjustments led to numerical improvements, the resulting reconstructions lacked the necessary detail for accurate approximation of the HRTF function, which is crucial for our purposes. For brevity, we will not present de-

Subject	Ear	Full Spectrum	Below 11kHz
Test	Left	5.325	3.857
	Right	5.128	4.000
Training	Left	4.788	3.925
	Right	4.724	3.914

Figure 3.5.3: LSD Results for dropout 10% model

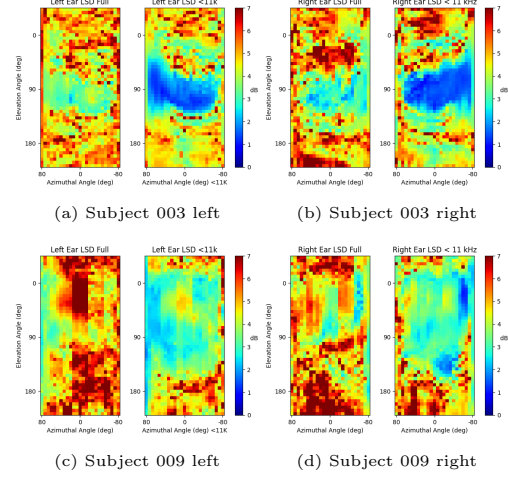


Figure 3.5.4: HRTF LSD images for subject 003 and 009 for the model with dropout rate 10% .

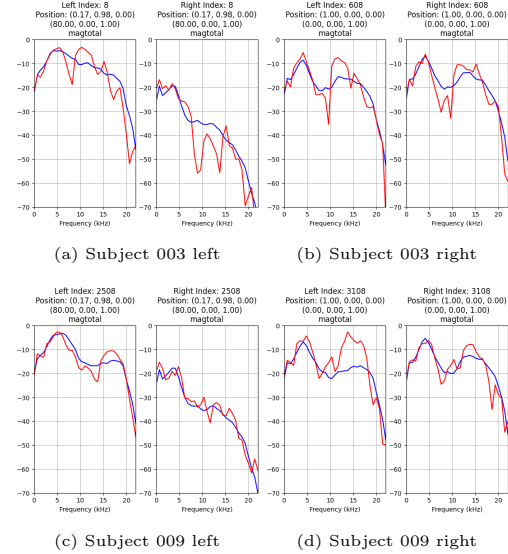


Figure 3.5.5: HRTF curves for a few positions for subject 003 (training) and 009 (test) for the model with dropout rate 10%.

tailed figures for these results; instead, we will report only the LSD numbers for test subject 009 in the Table 3.5.2.

Dropout Rate	Subject	Ear	LSD full	LSD <11kHz
1%	Test 009	Left	5.971	4.107
		Right	4.987	3.766
5%	Test 009	Left	5.393	3.575
		Right	5.306	3.544
20%	Test 009	Left	5.500	3.94
		Right	5.521	4.11

Table 3.5.2: LSD Results for Various Dropout Levels

3.5.3 Learning Rate

Another avenue of our experimentation was focused on adjusting the learning rate, incorporating its scheduling, and implementing early stopping. We employed a scheduler that reduced the learning rate by half whenever there was no improvement in validation loss. Additionally, we experimented with both small and large learning rates. For the sake brevity, we will not provide detailed figures from these experiments, as we believe they did not yield any significant, valuable improvements or new insights into solving the problem.

The results of these experiments are detailed in Table 3.5.3. As indicated, we practically did not benefit from implementing learning rate scheduling, and introducing a higher learning rate of $1e-3$ resulted in even more overfitting compared to $5e-4$, the rate used for our base model. We consider a learning rate of $5e-4$ to be optimal, and we believe the Adam optimizer, which we utilized for training, adequately handles basic

Learnng Rate	Subject	Ear	LSD Full	LSD <11kHz
lr-scheduler	Test	Left	5.782	4.253
		Right	5.025	3.961
	Training	Left	4.644	3.625
		Right	4.417	3.425
lr=1e-3	Test	Left	6.213	4.165
		Right	6.046	4.92
	Training	Left	2.713	2.049
		Right	3.201	2.482

Table 3.5.3: LSD Results for Various Dropout Levels

sic scheduling and necessary adjustments. Given that we are not dealing with complicated

models or fine-tuning, the Adam optimizer effectively manages the learning rate for our model.

3.5.4 Remove Mean

We normalized the HRTF data by removing the mean calculated on training data from every frequency bin, aiming to simplify the model’s task by working with a smaller range normalized data. This approach was inspired by the method introduced in [47].

However, we found that removing the mean did not benefit the model. The results are presented in Table 3.5.4, which details the numbers associated with this experiment.

Subject	Ear	Full Spectrum	Below 11kHz
Test	Left	5.552	3.896
	Right	6.098	5.434
Training	Left	3.053	2.270
	Right	2.769	2.086

Table 3.5.4: LSD Results for Mean Remove

3.5.5 One Model

The original paper (see Section 3.4.2) presented a methodology involving intermediate steps of model training for different aspects of the final HRTFs, followed by stacking trained networks before training the final model. In line with our focus on fully connected networks, we tested the hypothesis of the efficiency of this method against training only one fully connected network for matching anthropometric data to HRTFs. Table 3.5.5 presents the results of the above experiment, while Figure 3.5.7 displays the LSD images for frequencies. As before, we show numbers and graphs for subject 003 from the training dataset against subject 009 from the test dataset.

Subject	Ear	Full Spectrum	Below 11kHz
Test	Left	5.837	3.652
	Right	5.863	4.012
Training	Left	3.556	2.746
	Right	3.567	2.702

Table 3.5.5: LSD Results for One Model

As we can observe, there is a small improvement for LSD values for frequencies below 11kHz with one model paradigm compared to base model, which is clearly observed in Figure 3.5.7. We investigated the possibility of "average curve" problem that we encountered case of dropout, but upon examining the individual HRTF plots, it is evident that although the reconstruction is not perfect, the curves do not lack detail. We conclude that the two methods used in the Base model and One model are roughly equivalent in their performance. The only difference lies in efficiency; the proposed method from the original paper takes more than 12 hours to train as it trains 12 different models in a row stacking them to each other to construct final model, while the One Model paradigm takes around 1 hour. Some subsequent experiments were

conducted with the One Model approach aim to save time and resources, which will be presented in the following sections.

3.5.6 Masking and Weighting

In the series of following experiments, we aimed to test the hypothesis that the proposed network, or fully connected networks generally in this regard, could address the given problem by focusing on a smaller subset of the problem. This was achieved by limiting data position

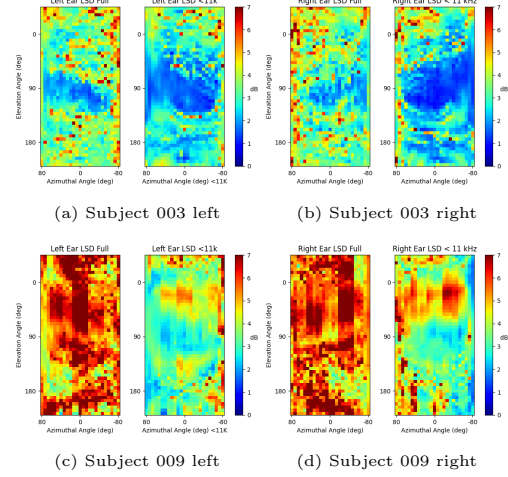


Figure 3.5.6: HRTF LSD images for subject 003 and 009 for the One Model.

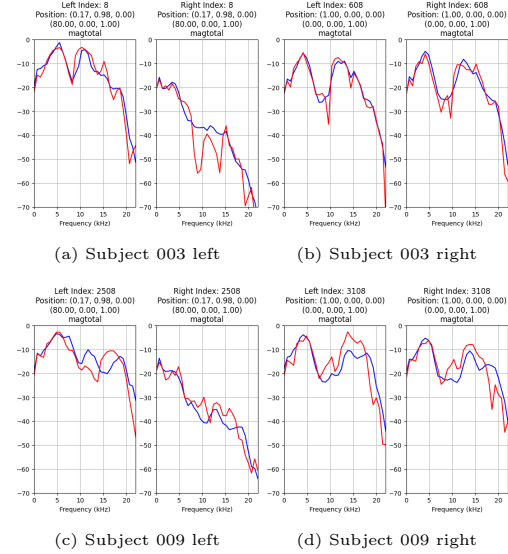


Figure 3.5.7: HRTF curves for a few positions for subject 003 (training) and 009 (test) for One Model.

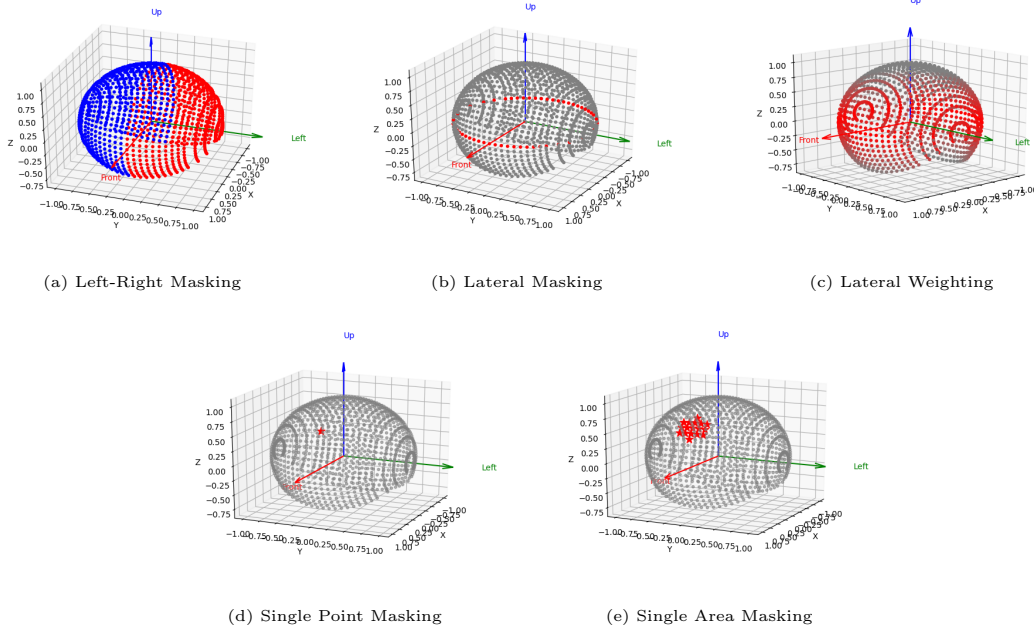


Figure 3.5.8: Different variations of masking and weighting methods for CIPIC data positions.

in the dataset or directing the model’s attention to a subset by applying weighting or the masking to loss function of the model during training. We conducted several experiments that explored such methodologies:

Lateral Masking: In this experiment, we masked all data points in the CIPIC dataset except for those at elevation zero. The goal was to assess whether our model could learn HRTF reconstruction in 2D space. Graphically, this masking procedure is shown in Figure 3.5.8b, where positions in red represent the only positions the model was trained on. Only positions in red were given loss weight of 1, and all other positions were given a weight of 0.

Lateral Weighting: This experiment serves as a more generalized version of the previous one. Instead of training the model solely on a narrow stripe at the no-elevation position, we assigned a weight to the loss function that is 1 at elevation zero and decreases inversely proportionally to the elevation, reaching 0 at the highest elevation point of the CIPIC data sphere. This approach is depicted in Figure 3.5.8c, where the intensity of the red corresponds to the weight given to the loss function.

Left-Right Masking: This experiment divides the CIPIC sphere into two halves, and

the model’s loss function is masked accordingly. The left outputs of the model calculate their loss considering only left positions, and the right outputs consider only the right side of the CIPIC sphere. This setup is illustrated in Figure 3.5.8a, where red and blue correspond to the left and right loss weights, respectively. This experiment tried to test if the models can divide their leaning capacity in two and learn data in such way.

Left-Right Weighting: This experiment is a generalized version of the previous one, where, instead of abruptly masking, we assign the most leftward position a loss weight of 1 for the left output of the network and decrease it linearly to the right, approximating 0 at the rightmost position. The same is done but vice versa for the right output of the model. This experiment aimed to test whether the model can focus the attention of its outputs to the corresponding sides and thereby improve generation quality. Additionally, we conducted a reversed version of this experiment where the left output of the model has a loss weight of 1 on the right side of the CIPIC sphere. This was done to evaluate if the model can focus its attention on the harder tasks of leaning farther positions and to explore the possibility of merging the solutions of both models if they prove successful.

Single Point Masking: This experiment tests the minimal hypothesis of whether the model can learn at least one position for all subjects and precisely generate a new point for test subjects given only anthropometric data, with position data fixed for all subjects. This concept is depicted in Figure 3.5.8d, where only the front upper position in front of the human subject receives a weight of 1 and actually participates in loss calculation and model training. This experiment effectively reduces the practical data size to 35 entries. We tested this approach with different positions in space, including left, right, and center at non-zero elevations.

Single Area Masking: This experiment represents an expanded version of its predecessor, concentrating the model’s learning efforts on a specific sector of the CIPIC sphere. The objective was to evaluate the model’s proficiency in mastering a discrete segment of the task, with the broader goal of employing multiple models in tandem should this strategy

prove successful. The design of one of the examples of this experiment is illustrated in Figure 3.5.8e, where only a small area centered in front and above the human subject is selected with a tolerance of 0.2 from the center. Only positions with a vector distance of less than 0.2 from the center target point are assigned a loss weight of 1, while all other positions are excluded from training by setting their loss to 0.

For the sake of convenience, We report the results of the aforementioned experiments by displaying LSD numbers and LSD images solely for the test subject 009. Table 3.5.6 and Figure 3.5.9 showcase these results. The inclusion of LSD images is particularly advantageous as full LSD numbers may not provide informative insights for models trained only on specific areas of points. Therefore, focusing on LSD values within the trained and attended regions is crucial for a comprehensive understanding.

To interpret the results of the experiments, let us examine Table 3.5.6 and Figure 3.5.9 systematically from top to bottom and left to right.

In the first experiment, Lateral Masking, despite being trained on only 50 data points (out of a total of 1250 positions), the model shows promising results. However, it falls short of its primary objective to maintain a low LSD score along the zero-elevation surface. This shortcoming is evident in the LSD images, where, contrary to the anticipated green or blue areas around zero elevation horizontal stripe indicating lower error, we observe red areas denoting higher error in both the left and right ears images. This discrepancy underscores the model’s inability to sustain an LSD below 4 dB, even for the simplified task of focusing

Experiment	Subject	Ear	LSD full	LSD <11kHz
Lateral Masking	Test 009	Left	5.223	4.298
		Right	5.479	4.669
Lateral Weighting	Test 009	Left	6.077	4.367
		Right	5.583	4.114
Left-Right Masking	Test 009	Left	6.945	4.828
		Right	6.348	4.166
Left-Right Weighting	Test 009	Left	6.016	4.356
		Right	5.349	3.734
Left-Right Weighting Reversed	Test 009	Left	5.813	4.016
		Right	5.161	4.089
Single Point Masking ($\theta = 52$, $\phi = 63$)	Test 009	Left	10.988 ¹	9.150 ¹
		Right	12.693 ¹	9.827 ¹
Single Area Masking ($\theta = 52$, $\phi = 63$, $\tau = 0.2$)	Test 009	Left	8.722 ¹	7.275 ¹
		Right	12.452 ¹	9.472 ¹

Table 3.5.6: LSD Results for Various Masking and Weighting Strategies

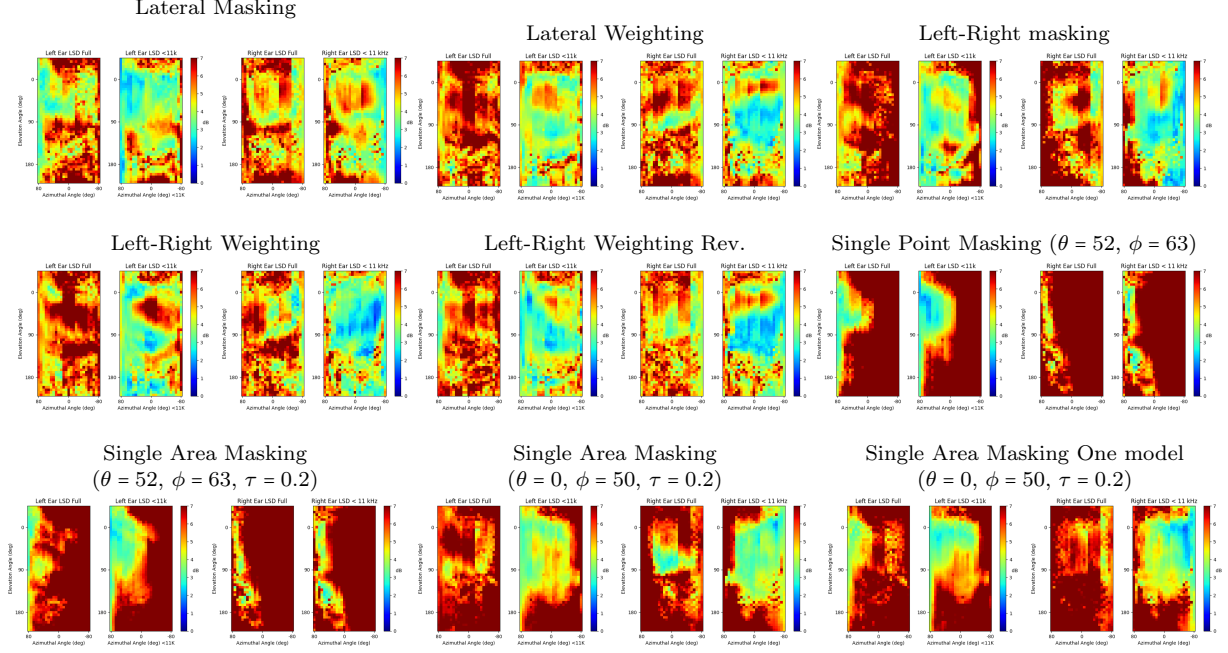


Figure 3.5.9: LSD Images for Various Masking and Weighting Strategies

on the no-elevation positions it was trained on.

Proceeding further, we encounter the Lateral Weighting experiment, which proved to be less effective than the baseline model, not only in reconstructing the zero-elevation stripe but also positions above and below it. Intriguingly, it is evident that the previous model, which focused exclusively on zero-elevation positions, predicted some areas in front and above the human subject (0-degree to 90-degree area in the image) more accurately than this one.

Next, we examine the Left-Right Masking, which also proved to be substantially inefficient. Essentially, the model fails to predict HRTFs for the positions opposite to given ear effectively, which aligns with the limitations imposed by the Masking constraint. However, surprisingly, the model does not show improvement even in the areas where its focus was directed, still exhibiting numerous red areas, indicating high errors even there.

The Left-Right Weighting and Left-Right Weighting Reversed exhibited significantly better results than the other experiments in this set and almost mirrored the numbers and images produced by the base model. However, we did not observe the anticipated outcome where Left-Right Weighting would display more green and blue areas, indicating superior

reconstruction in the azimuth space from 80 to 0 degrees on the left and 0 to -80 degrees on the right side. Conversely, Left-Right Weighting Reversed was expected to exhibit the effect in the opposite direction. This could have allowed us to divide the task of HRTF estimation between two parallel models, predicting the left ear HRTF from the model that assigned more weight to the corresponding side of ear position, and predicting the opposite ear HRTF from a model that placed more weight on the opposite side. This outcome suggests that the model does not benefit from decomposing the task, pointing to a more fundamental issue. We hypothesize that this could be related to the quantity and significance of the data used for prediction, as well as the meaningfulness of the parameters for data collection, indicating deeper underlying problems beyond the design of the ML model itself.

To further explore the task of breaking down the problem into smaller sub tasks and testing the model’s capacity to solve smaller, partial problems—which could later aggregate into insights for a more comprehensive solution—we evaluated the model’s ability to accurately predict a single point. The selected point was position index 165, corresponding to a left side azimuth of 52 degrees and an elevation angle of 63 degrees. While the reconstruction for the left ear was not perfect, it was somewhat tolerable; however, the model entirely failed to accurately predict for the right ear’s HRTF. Additionally, an interesting observation from the single-point experiments was that, regrettably, the model does not differentiate between subjects, predicting an average curve that seems to minimize loss effectively for the small dataset we are getting after focusing on one point. This phenomenon might be attributed to so-called degenerate solutions problem in ML, where the model converges to a trivial solution. We believe this could be a result of the dataset being either too small or not representative of the underlying data distribution, leading the model to fail in learning meaningful patterns and instead resort to generating constant predictions that satisfies the local minima of loss function. In this case, the dataset size is 31 items in training+validation set, with input sizes of 17+10 and an output size of 64. We suspect this might be a reason for the problem.

To address the problem of an overly small dataset and to render the model more meaningful, we experimented with directing the model’s full attention to a small area of data point positions instead of one point. This approach seemed logical not only because of intuitions from physics and the general perception that sound from areas closer to each other tends to be similar but also because it was substantiated by the original paper and numerous other studies. These sources indicate there is always a strong correlation between neighboring positions in the dataset, between which interpolation and extrapolation algorithms have performed well. We hypothesized that a group of data points would provide the model with enough information for effective training, enabling it to learn the defining forces in anthropometric data. We took the same data point of index 165, and instead of focusing on a single point, we introduced a tolerance area with $\tau = 0.2$, where every position within a vector distance less than 0.2 from the center point would be included in the area. Unfortunately, the model again proved inefficient, failing to reconstruct the HRTF data for the opposing ear accurately. To further test this hypothesis, we additionally explored the model’s performance on several different positions and observed that the model does not efficiently learn HRTF curves. We present data point 615, which corresponds to a mid-front position with an elevation angle of 50 degrees relative to the human subject. As observed in the images, the model fails to accurately predict this area for the left ear while performing relatively well for the right ear.

Lastly, for the sake of experiment, we trained the One Model variant on the same single area task, to determine its capability to learn the front upper positional area (position index 615). As illustrated in the last image on the bottom right side of Figure 3.5.9, the model exhibits no improvement over the base model that was focused on the same area.

In conclusion, our series of experiments on HRTF prediction models revealed limitations in accuracy across various approaches, including lateral masking and weighting, and focused attention on specific data points or areas. There were no improvements and Model did not behave in the way we imagined they would. These findings suggest the existence of some

fundamental problems in the task of predicting HRTF data from anthropometrics, at least with CIPIC dataset and the given setting.

3.5.7 Cleaned and Smoothed CIPIC

The CIPIC dataset was found to have several problems by different generations of researchers at UC San Diego while conducting research and working on [65] and its continuation, of which this chapter is also a part. Several problems were identified, such as inconsistencies in HRTF measurements, unexpected jumps in some frequency bins, and most importantly, the "5th ring problem," which showed systematic inconsistencies in

Experiment	Subject	Ear	LSD full	LSD <11kHz
notch-smoothing-1	Test 009	Left	5.616	3.745
		Right	5.421	3.987
notch-smoothing-2	Test 009	Left	5.111	3.652
		Right	5.511	3.884
notch-smoothing-3	Test 009	Left	5.387	3.834
		Right	5.579	4.226
5thring-6	Test 009	Left	5.405	3.854
		Right	5.243	4.147

Table 3.5.7: LSD Results for Various Masking and Weighting Strategies

the 5th ring of the CIPIC spherical ring. This issue was attributed to the problem of a changed volume of the speaker responsible for the 5th ring. These problems were addressed, and the CIPIC data was cleaned and smoothed in several ways, producing four different versions of the CIPIC data: "notch-smoothing-1," "notch-smoothing-2," "notch-smoothing-3," and "5thring-6," with the latter specifically addressing the "5th ring problem." We will not go through the details of smoothing and cleaning the CIPIC data as this was not done by the author of this chapter, and there is a lack of good documentation on the exact procedures.

We conducted experiments and tried to train our model with all the types of cleaned and smoothed CIPIC data. The results of the experiments are given in Table 3.5.7. Unfortunately, as observed in the table, the cleaned and smoothed CIPIC data, in any of its iterations, does not provide any substantial improvement to the model. This suggests that the problem is not simply the cleanliness or smoothness of the data but may be more fundamental to the ML modeling of CIPIC data.

3.6 Conclusion

In this chapter, we explored the theory and practice behind spatial sound, focusing on Head-Related Transfer Functions (HRTFs) and their modeling and personalization using ML. We reviewed the theoretical background of HRTFs and discussed the challenges in the spatial audio field, particularly regarding HRTF acquisition and synthesis using ML-based methods. We detailed the measurement process and the challenges of acquiring accurate HRTF data, which have driven the development of ML models in this field. We presented existing datasets and methods for evaluating simulated HRTFs. Following this, we reviewed existing literature and methodologies for HRTF generation with ML methods. Finally, we detailed our attempts to tackle the problem of personalizing HRTFs with ML through various methodologies, addressing the challenges of overfitting and generalization in small datasets like CIPIC.

Our exploration of the literature and our own experiments revealed that, despite tremendous amount of work and effort directed to the field, the problem of generating HRTFs from anthropometric data still remains largely unresolved. We believe, the key issues of the field lie in the lack of extensive datasets and the lack of precision of existing measurement processes. We found that most existing work, especially ones working with current datasets, do not produce satisfactory results, even when using state-of-the-art ML models and methods. This indicates that current datasets, like CIPIC, are not sufficient for training robust ML models for HRTF personalization. Similarly, despite various approaches we tried, our model failed to generalize effectively and generate satisfactory results. To our qualified satisfaction, we observed that our results fall within the same range of LSD error as those from other more recent and sophisticated methods. While it is reassuring that our approach is somewhat reasonable, it is also disappointing as it once again proves that the problem is generally unresolved. Moreover, our exploration suggests that beyond dataset limitations, there might be fundamental aspects of HRTFs that are not yet fully understood, preventing ML methodologies from modeling them correctly. The absence of standardized methods for

measurements, the lack of a precise list of necessary anthropometric data, and other factors possibly add to the complexity of HRTF modeling. These insights point towards the need for more comprehensive datasets, better understanding of the theory, and standardized measurement processes, which could form the basis for future research in this challenging but exciting field.